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1 ABSTRACT

1.1 Objective

A Packed column was applied in mass transfer processes such as gas absorption, distillation, and liquid extraction. These experiments were to investigate the hydrodynamic characteristics and the effects of the gas and liquid velocities on the operating performance of the packed column. In the packed column, the inlet liquid is distributed over the top of the packed section and flows downward in the column, while the inlet gas was entered the distributing space below the packed section and flows upward through the interstices in the packing countercurrent to the flow of the liquid.

Through the experiments, the students could determine:

1. The effects of gas and liquid velocities on the pressure drop of gas flow in the packed column.
2. The variation of the friction coefficient of dried packing (f_{dry}) as a function of Reynolds number (Re) of inlet gas flow rate and inferring the experimental relations.
3. The change of " σ " factor, showing the relationship between the pressure drop of the gas flow through the dried packing and that through the wetted packing in different inlet liquid velocities.
4. The diagram of limiting operation performance of packed column (loading and flooding points).

1.2 Experiment Method

Gas flow is supplied with different flowrates to the packed column containing ceramic chocks. Pressure drop is surveyed in turn when only gas flow goes into the column (dried column) and when gas flow goes into the column combined with liquid flow going down from the top with different flowrates (wetted column).

1.3 Result Data

The following table presents the raw experimental pressure drop measurements (ΔP in mmH_2O) for various gas flow percentages and liquid irrigation rates (L in gal/min).

Table 1: Raw experimental data (ΔP in mmH_2O)

$G[\%]$	$L = 0$	$L = 0.2$	$L = 0.4$	$L = 0.6$	$L = 0.8$	$L = 1.0$	$L = 1.2$	$L = 1.4$	$L = 1.6$
10	1	1	1	0	2	3	3	5	8
20	5	6	8	4	8	9	14	20	25
30	10	12	16	17	20	23	35	62	78
40	19	22	26	31	53	49	75	118	142
50	29	32	42	49	107	95	163	328	258
60	38	46	62	95	164	180	246	—	—
70	52	62	109	174	348	303	332	—	—
80	70	82	172	295	412	393	418	—	—
90	90	108	267	377	—	—	—	—	—
100	108	141	315	434	—	—	—	—	—

2 THEORY

2.1 Pressure drop of inlet gas flow

The pressure drop (ΔP_{dry}) of the gas flow through the packed column depends on the gas velocity (G) in the condition of dried packing (no countercurrent liquid flow). The pressure drop in the packed column increases as the velocity of the inlet gas flowing through interstices in the packing materials is raised. The pressure drop increases with the 1.8 – 2.0 power of the velocity of the gas:

$$\Delta P_{dry} = \alpha G^n \quad (1)$$

Where $n = 1.8 - 2.0$.

When liquid enters the column at the top of the packing and flows downward, the interstices in the packing are reduced due to the occupation of the liquid in the free volume of the packing material. Therefore, the displacement of the gas flow becomes harder. With an increment of gas flow rate, the impedimental effect of the liquid flow regularly increases. When the gas flow rate reaches a critical value, the pressure drop rises gradually. If the gas flow rate exceeds this loading point, the pressure drop increases more rapidly and does not follow Equation (1). In the local regions of the column, the liquid becomes a continuous phase, and the column reaches the flooding point. The entire column may fill with fluid at this flooding point.

2.2 The friction factor f_{dry} as a function of Re_c for dried column

The pressure drop (ΔP_{dry}) can be determined using the Ergun equation:

$$\frac{\Delta P_{dry}}{Z} \frac{\varepsilon^3}{(1 - \varepsilon)} \frac{d_{td} \rho_G}{G^2} = \frac{150(1 - \varepsilon)}{Re} + 1.75 \quad (2)$$

Chilton and Colburn suggested a relation:

$$\Delta P_{dry} = 2 f_{dry} \frac{G^2 Z}{\rho_g D_h} \gamma_h \gamma_w \quad [N/m^2] \quad (3)$$

Where:

- Z : packing depth [m]
- G : gas flow velocity per unit of packed section [$\text{kg}/\text{s} \cdot \text{m}^2$]
- D_h : nominal size of packing material [m]
- ρ_g : density of gas flow [kg/m^3]
- γ_h : adjustment factor of packing material (0.35 for Raschig rings)
- γ_w : adjustment factor for wall effect (1.0)

Zhavoronkov suggested another relation:

$$\Delta P_{dry} = \frac{2f_{dry}G^2Z}{\varepsilon^2\rho_G D_e} \quad [N/\text{m}^2] \quad (4)$$

Where:

- ε : porosity of packing material
- $D_e = 4\varepsilon/a$: equivalent diameter of packing material [m]
- a : specific surface area of packing material [m^2/m^3]

Friction factor f_{dry} is a function of dimensionless Re_c :

$$Re_c = \frac{GD_e}{\varepsilon\mu} = \frac{4G}{a\mu} \quad (5)$$

Where μ : viscosity of gas flow [kg/ms].

In the turbulent flow region ($50 < Re_c < 7000$):

$$f_{dry} = \frac{3.8}{Re_c^{0.2}} \quad (6)$$

2.3 Pressure drop ΔP_{wet} for wetted column

The correlation between ΔP_{dry} and ΔP_{wet} is:

$$\Delta P_{wet} = \sigma \Delta P_{dry} \implies f_{wet} = \sigma f_{dry} \quad (7)$$

Leva suggested a relation for σ as a function of liquid velocity L [$\text{kg}/\text{m}^2\text{s}$]:

$$\sigma = 10^{\Omega L} \quad \text{or} \quad \log \sigma = \Omega L \quad (8)$$

For 12.7 mm Raschig rings (random packing, $\varepsilon = 0.568$, $L = 0.39 - 11.7 \text{ kg}/\text{m}^2\text{s}$), $\Omega = 0.084$.

From previous studies, the relation between the ratio of $\Delta P_{wet}/\Delta P_{dry}$ and the coefficient of fluid irrigation (A) is:

$$A = \sqrt{\frac{1.75}{Re_L} \left(\frac{G_L}{F \rho_L} \right) \frac{q}{2g\varepsilon^2}} \quad (9)$$

For $A < 0.3$ and ceramic packing material with $d < 30$ mm:

$$\frac{\Delta P_{wet}}{\Delta P_{dry}} = \frac{1}{(1 - A)^3} \quad (10)$$

Where $Re_L = \frac{4G_L}{Fa\mu_L}$.

2.4 The flooding point of packed column

Flooding occurs when the liquid seizes the packed section entirely. Zhavoronkov defined the flooding situation using two dimensionless numbers π_1 and π_2 :

$$\pi_1 = \left(\frac{f_{dry}a}{\varepsilon^3} \right) \frac{v^2}{2g} \cdot \frac{\rho_G}{\rho_L} \cdot \mu_{td}^{0.2} \quad (11)$$

$$\pi_2 = \frac{L}{G} \sqrt{\frac{\rho_G}{\rho_L}} \quad (12)$$

Where:

- v : gas flow rate before feeding [m/s]
- $\mu_{td} = \mu_l/\mu_{water}$: relative viscosity

3 EQUIPMENT

3.1 Diagram of experimental apparatus

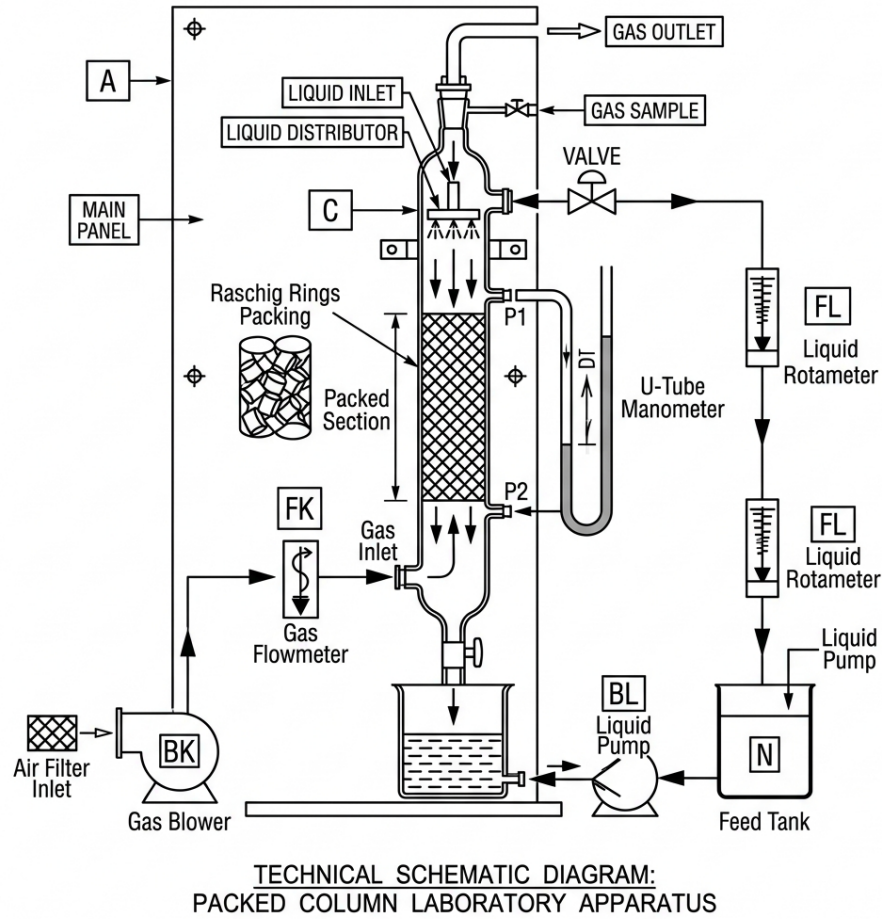


Figure 1: Diagram of experimental apparatus

The experimental unit consists of:

1. A glass column where Raschig rings are packed randomly.
2. Gas feeding equipment: Air blower (B_K), gas pipe, U-tube manometer, differential pressure gauge, and gas flowmeter (scale 8-100%, $V = 0.286 \text{ m}^3/\text{min}$).
3. Water feeding equipment: Feed tank (N), water pump (B_L), and liquid flowmeter (range 0-1.6, $G_L = 5.805 \text{ l/min}$).

3.2 Characteristics of packed column

Glass column:

- Diameter: $d = 0.09 \text{ m}$

- Height: $H = 0.805$ m
- Packing height: $Z = 0.42$ m

Packing material (Raschig rings):

- Nominal diameter: $d = 12.7$ mm
- Specific surface area: $a = 370 - 380$ m^2/m^3
- Porosity: $\varepsilon = 0.586$

3.3 List of components

- **A**: Panel
- **C**: Packed column
- **E**: Bottom part of column
- F_K : Gas flowmeter
- **g**: Liquid level device (pipe) at the bottom
- F_L : Liquid flowmeter
- B_K : Air blower (1.0 Hp)
- B_L : Liquid pump (0.5 Hp)
- **N**: Liquid tank
- **AK**: Pressure gauge

4 METHOD

4.1 Preparation

1. Close all liquid valves.
2. Turn on the air valve and switch on the air blower for 5 minutes to remove moisture from the column. Then, shut down the blower.
3. Open valve "1" and valve "3", and pump water into the bottom part of the column.
4. Use valve "2" to adjust the water level to equal the height of the level device (approximately 3/4 of the bottom part).
5. Turn off the pump and close valve "3".

4.2 Measurement of pressure drop for dried column

1. Ensure all liquid valves are closed.
2. Open air valve "6" while valve "5" is closed.
3. Turn on the air blower and open valve "5" slowly to adjust the gas flow.
4. For a fixed gas flow rate, determine ΔP_{dry} on the pressure gauge (in mmH_2O).
5. Turn off the blower and wait for 5 minutes between measurements.

4.3 Measurement for pressure drop of the wetted column

1. Switch on the air blower and adjust gas flow rate to 15-20% of maximum.
2. Open liquid valve "1" and turn on the pump.
3. Use valve " V_L " from the liquid flowmeter to adjust the water flow rate.
4. At a specific liquid flow rate, alter the air flow rate and measure ΔP_{wet} until the column reaches the flooding point.
5. Repeat the process for different liquid flow rates.

Note: Maintain the water level at 3/4 of the bottom part throughout the measurement using valve "4" or "2" as needed.

5 RESULTS

5.1 Experimental Data and Calculations

The experimental results for various gas flow rates ($G\%$) and liquid irrigation rates (L) are presented below. The pressure drop per unit height ($\Delta P/Z$), friction factor (f), Reynolds number (Re), and the σ factor were calculated for each data point.

Table 2: Experimental results for $L = 0.0$ gal/min

$G[\%]$	$G[kg/s.m^2]$	$\Delta P[mmH_2O]$	$\Delta P/Z[N/m^3]$	σ	f_{wet}	Re
10	0.0847	1.0	23.36	1.00	4.02	46.86
20	0.1694	5.0	116.79	1.00	5.02	93.73
30	0.2541	10.0	233.57	1.00	4.47	140.59
40	0.3388	19.0	443.79	1.00	4.79	187.45
50	0.4235	29.0	677.36	1.00	4.75	234.32
60	0.5083	38.0	887.57	1.00	4.31	281.18
70	0.5930	52.0	1214.57	1.00	4.28	328.04
80	0.6776	70.0	1635.00	1.00	4.41	374.91
90	0.7623	90.0	2102.14	1.00	4.48	421.77
100	0.8471	108.0	2522.57	1.00	4.35	468.63

Table 3: Experimental results for $L = 0.2$ gal/min

$G[\%]$	$G[kg/s.m^2]$	$\Delta P[mmH_2O]$	$\Delta P/Z[N/m^3]$	σ	f_{wet}	Re
10	0.0847	1.0	23.36	1.00	4.02	46.86
20	0.1694	6.0	140.14	1.20	6.02	93.73
30	0.2541	12.0	280.29	1.20	5.37	140.59
40	0.3388	22.0	513.86	1.16	5.54	187.45
50	0.4235	32.0	747.43	1.10	5.24	234.32
60	0.5083	46.0	1074.43	1.21	5.21	281.18
70	0.5930	62.0	1448.14	1.19	5.10	328.04
80	0.6776	82.0	1915.14	1.17	5.17	374.91
90	0.7623	108.0	2522.57	1.20	5.37	421.77
100	0.8471	141.0	3292.50	1.31	5.68	468.63

Table 4: Experimental results for $L = 0.4$ gal/min

$G[\%]$	$G[kg/s.m^2]$	$\Delta P[mmH_2O]$	$\Delta P/Z[N/m^3]$	σ	f_{wet}	Re
10	0.0847	1.0	23.36	1.00	4.02	46.86
20	0.1694	8.0	186.86	1.60	8.03	93.73
30	0.2541	16.0	373.71	1.60	7.15	140.59
40	0.3388	26.0	607.29	1.37	6.55	187.45
50	0.4235	42.0	980.93	1.45	6.87	234.32
60	0.5083	62.0	1448.14	1.63	7.03	281.18
70	0.5930	109.0	2545.93	2.10	8.97	328.04
80	0.6776	172.0	4017.43	2.46	10.84	374.91
90	0.7623	267.0	6236.14	2.97	13.28	421.77
100	0.8471	315.0	7357.50	2.92	12.69	468.63

Table 5: Experimental results for $L = 0.6$ gal/min

$G[\%]$	$G[kg/s.m^2]$	$\Delta P[mmH_2O]$	$\Delta P/Z[N/m^3]$	σ	f_{wet}	Re
10	0.0847	0.0	0.00	0.00	0.00	46.86
20	0.1694	4.0	93.43	0.80	4.02	93.73
30	0.2541	17.0	397.07	1.70	7.60	140.59
40	0.3388	31.0	724.07	1.63	7.81	187.45
50	0.4235	49.0	1144.50	1.69	8.02	234.32
60	0.5083	95.0	2218.93	2.50	10.77	281.18
70	0.5930	174.0	4064.14	3.35	14.33	328.04
80	0.6776	295.0	6890.36	4.21	18.59	374.91
90	0.7623	377.0	8805.43	4.19	18.75	421.77
100	0.8471	434.0	10136.57	4.02	17.48	468.63

Table 6: Experimental results for $L = 0.8$ gal/min

$G[\%]$	$G[kg/s.m^2]$	$\Delta P[mmH_2O]$	$\Delta P/Z[N/m^3]$	σ	f_{wet}	Re
10	0.0847	2.0	46.71	2.00	8.03	46.86
20	0.1694	8.0	186.86	1.60	8.03	93.73
30	0.2541	20.0	467.14	2.00	8.94	140.59
40	0.3388	53.0	1237.93	2.79	13.36	187.45
50	0.4235	107.0	2499.21	3.69	17.52	234.32
60	0.5083	164.0	3830.36	4.32	18.59	281.18
70	0.5930	348.0	8128.29	6.69	28.66	328.04
80	0.6776	412.0	9622.93	5.89	25.96	374.91

Table 7: Experimental results for $L = 1.0$ gal/min

$G[\%]$	$G[kg/s.m^2]$	$\Delta P[mmH_2O]$	$\Delta P/Z[N/m^3]$	σ	f_{wet}	Re
10	0.0847	3.0	70.07	3.00	12.05	46.86
20	0.1694	9.0	210.21	1.80	9.03	93.73
30	0.2541	23.0	537.21	2.30	10.28	140.59
40	0.3388	49.0	1144.50	2.58	12.35	187.45
50	0.4235	95.0	2218.93	3.28	15.55	234.32
60	0.5083	180.0	4204.29	4.74	20.40	281.18
70	0.5930	303.0	7077.21	5.83	24.95	328.04
80	0.6776	393.0	9179.14	5.61	24.77	374.91

Table 8: Experimental results for $L = 1.2$ gal/min

$G[\%]$	$G[kg/s.m^2]$	$\Delta P[mmH_2O]$	$\Delta P/Z[N/m^3]$	σ	f_{wet}	Re
10	0.0847	3.0	70.07	3.00	12.05	46.86
20	0.1694	14.0	327.00	2.80	14.05	93.73
30	0.2541	35.0	817.50	3.50	15.64	140.59
40	0.3388	75.0	1751.79	3.95	18.90	187.45
50	0.4235	163.0	3807.00	5.62	26.68	234.32
60	0.5083	246.0	5745.86	6.47	27.88	281.18
70	0.5930	332.0	7754.79	6.38	27.34	328.04
80	0.6776	418.0	9763.07	5.97	26.34	374.91

Table 9: Experimental results for $L = 1.4$ gal/min

$G[\%]$	$G[kg/s.m^2]$	$\Delta P[mmH_2O]$	$\Delta P/Z[N/m^3]$	σ	f_{wet}	Re
10	0.0847	5.0	116.79	5.00	20.09	46.86
20	0.1694	20.0	467.14	4.00	20.07	93.73
30	0.2541	62.0	1448.14	6.20	27.71	140.59
40	0.3388	118.0	2756.14	6.21	29.74	187.45
50	0.4235	328.0	7660.93	11.31	53.70	234.32

Table 10: Experimental results for $L = 1.6$ gal/min

$G[\%]$	$G[kg/s.m^2]$	$\Delta P[mmH_2O]$	$\Delta P/Z[N/m^3]$	σ	f_{wet}	Re
10	0.0847	8.0	186.86	8.00	32.14	46.86
20	0.1694	25.0	583.93	5.00	25.09	93.73
30	0.2541	78.0	1821.86	7.80	34.86	140.59
40	0.3388	142.0	3316.43	7.47	35.79	187.45
50	0.4235	258.0	6026.57	8.90	42.24	234.32

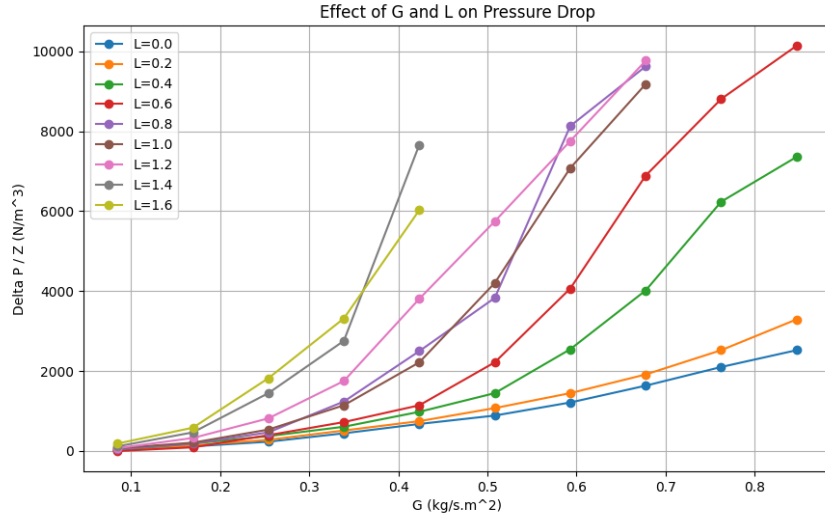


Figure 2: Effect of gas flow rate G and liquid irrigation rate L on the pressure drop $\Delta P/Z$.

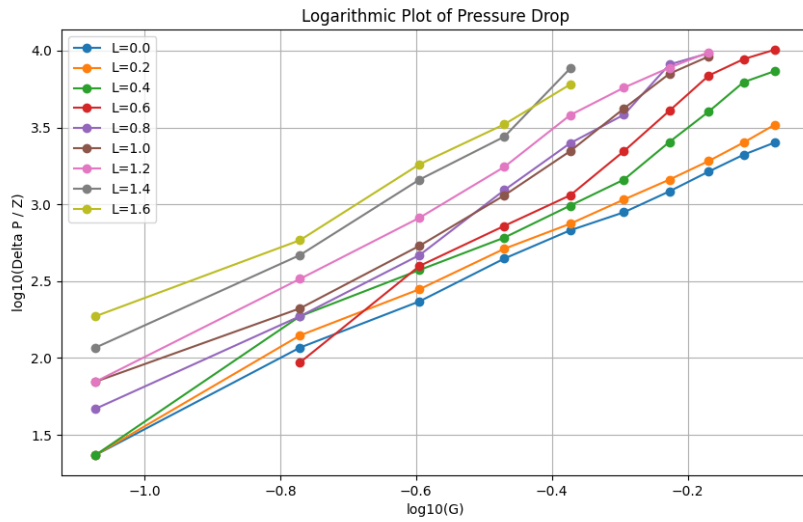


Figure 3: Logarithmic plot of pressure drop vs. gas flow rate.

5.2 Flooding Point Analysis

The flooding point was identified for each liquid irrigation rate. The dimensionless numbers π_1 and π_2 were calculated to determine the flooding limit of the packed column.

Table 11: Flooding point data for different liquid flow rates

$L[gal/min]$	$G^*[\%]$	$G^*[kg/s.m^2]$	$v[m/s]$	π_1	π_2	$\log \pi_1$	$\log \pi_2$
0.2	100	0.8471	0.7493	0.2583	0.0952	-0.5879	-1.0215
0.4	100	0.8471	0.7493	0.2583	0.1903	-0.5879	-0.7205
0.6	100	0.8471	0.7493	0.2583	0.2855	-0.5879	-0.5444
0.8	80	0.6776	0.5994	0.1674	0.4759	-0.7762	-0.3225
1.0	80	0.6776	0.5994	0.1674	0.5948	-0.7762	-0.2256
1.2	80	0.6776	0.5994	0.1674	0.7138	-0.7762	-0.1464
1.4	50	0.4235	0.3746	0.0694	1.3324	-1.1589	0.1246
1.6	50	0.4235	0.3746	0.0694	1.5227	-1.1589	0.1826

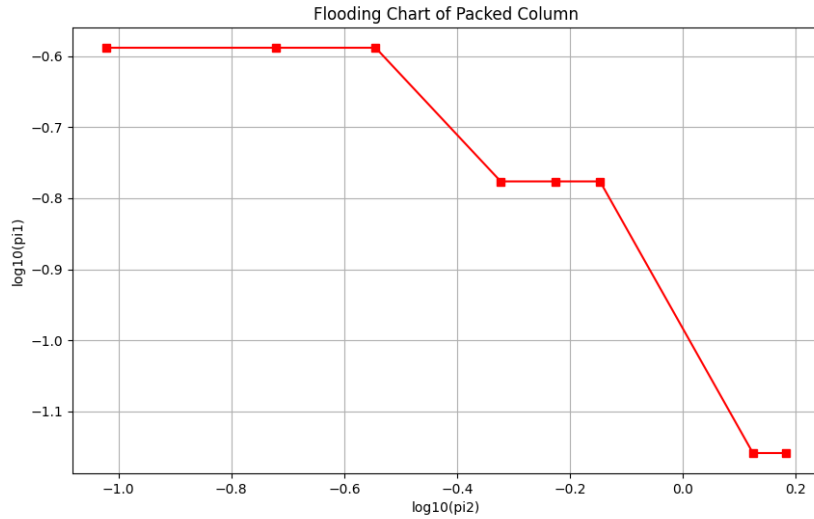


Figure 4: Flooding chart of the packed column ($\log \pi_1$ vs. $\log \pi_2$).

5.3 Empirical Results and Equations

The relationship between the hydrodynamic parameters was analyzed using regression analysis. Table 12 summarizes the empirical equations derived from the experimental data.

Table 12: The empirical results and equations

Relation	Empirical Equation
$\Delta P_{ck}/Z$ vs G at $L = 0$	$y = 3070.4x^2 + 359.51x - 8.9536$
$\Delta P_{c\ddot{o}}/Z$ vs G at $L = 0.2$	$y = 3723.9x^2 - 28.828x + 56.836$
$\Delta P_{c\ddot{o}}/Z$ vs G at $L = 0.4$	$y = 4784.4x^2 - 30.917x + 28.029$
$\Delta P_{c\ddot{o}}/Z$ vs G at $L = 0.6$	$y = 9146x^2 - 1549.4x + 184.35$
$\Delta P_{c\ddot{o}}/Z$ vs G at $L = 0.8$	$y = 17323x^2 - 4402.1x + 397.07$
$\Delta P_{c\ddot{o}}/Z$ vs G at $L = 1$	$y = 11702x^2 - 1448.7x + 109.08$
$\Delta P_{c\ddot{o}}/Z$ vs G at $L = 1.2$	$y = 35809x^2 - 6342.2x + 385.39$
$\Delta P_{c\ddot{o}}/Z$ vs G at $L = 1.4$	$y = 35809x^2 - 6342.2x + 385.39$
$\Delta P_{c\ddot{o}}/Z$ vs G at $L = 1.6$	$y = 23601x^2 - 3240x + 198.54$
$\log f_{ck}$ vs Re_{ck} at $L = 0$	$y = -0.0006x + 0.8656$
$\log f_{c\ddot{o}}$ vs Re_{ck} at $L = 0.2$	$y = -0.0007x + 0.9235$
$\log f_{c\ddot{o}}$ vs Re_{ck} at $L = 0.4$	$y = -0.0006x + 0.9861$
$\log f_{c\ddot{o}}$ vs Re_{ck} at $L = 0.6$	$y = -0.0005x + 1.046$
$\log f_{c\ddot{o}}$ vs Re_{ck} at $L = 0.8$	$y = -0.0002x + 1.0543$
$\log f_{c\ddot{o}}$ vs Re_{ck} at $L = 1$	$y = 0.0003x + 1.1032$
$\log f_{c\ddot{o}}$ vs Re_{ck} at $L = 1.2$	$y = 0.0011x + 1.0434$
$\log f_{c\ddot{o}}$ vs Re_{ck} at $L = 1.4$	$y = 0.0015x + 1.0995$
$\log f_{c\ddot{o}}$ vs Re_{ck} at $L = 1.6$	$y = 0.002x + 1.1082$
$\log \sigma$ vs L at $G = 10\%$	$y = 0.13x - 0.001$
$\log \sigma$ vs L at $G = 20\%$	$y = 0.38x + 0.02$
$\log \sigma$ vs L at $G = 30\%$	$y = 0.42x - 0.02$

6 DISCUSSION

1. **Pressure Drop Trends:** Based on the experimental data and charts:
 - **Dry Column ($L = 0$):** As gas flow rate G increases, the pressure drop ΔP increases approximately linearly on a log-log scale, following the theoretical prediction $\Delta P \propto G^n$.
 - **Wetted Column ($L > 0$):** As G increases, ΔP increases more rapidly than in the dry case. As the liquid irrigation rate L increases, the column reaches the flooding point at lower gas velocities. This is because the liquid occupies the void space, increasing the actual gas velocity and resistance.
2. **Friction Factor and Reynolds Number:** The friction factor f was found to decrease as the Reynolds number Re increases, which is typical for turbulent flow in packed beds. The chart of f vs. Re helps determine the optimal operating range to maximize mass transfer efficiency while maintaining a manageable pressure drop.
3. **Comparison with Predictions:** The relationship between $\log(\Delta P/Z)$ and $\log G$ is linear as predicted. For wetted columns, two distinct regions are observed: below and above the loading point. Below the loading point, ΔP increases gradually. Above the loading point, ΔP rises sharply as the column approaches flooding.

4. **Sources of Error:** Several factors could contribute to discrepancies between experimental results and theoretical predictions:

- **Flow Instability:** The liquid pump and the air blower (fan) may produce unstable flow rates, leading to fluctuating pressure readings.
- **Bottom Water Level:** If the water column maintained at the bottom of the column is insufficient, it can lead to flooding in the differential pressure tube, affecting the results.
- **Measurement Errors:** Human error in reading the manometer scales and flowmeters, especially under fluctuating conditions.
- **Equilibrium Time:** Not allowing sufficient time (suggested 10 minutes) for the system to stabilize after adjusting flow rates.
- **Ambient Conditions:** Variations in temperature and pressure during the experiment affecting air density and viscosity.

7 CALCULATION

7.1 Sample Calculation for $L = 0$ and $G = 10\%$

1. Gas mass velocity (G):

$$G = \frac{G\% \cdot \rho_G \cdot V_{max}}{60 \cdot F} = \frac{10 \cdot 1.1305 \cdot 0.286}{60 \cdot 0.0064 \cdot 100} = 0.0847 \text{ kg/s.m}^2 \quad (13)$$

2. Pressure drop ($\Delta P/Z$): For $\Delta P = 1 \text{ mmH}_2\text{O}$:

$$\Delta P[N/m^2] = \Delta P_{mmH_2O} \cdot g = 1 \cdot 9.81 = 9.81 \text{ N/m}^2 \quad (14)$$

$$\frac{\Delta P}{Z} = \frac{9.81}{0.42} = 23.36 \text{ N/m}^3 \quad (15)$$

3. Reynolds number (Re):

$$Re = \frac{4G}{a\mu} = \frac{4 \cdot 0.0847}{375 \cdot 1928 \cdot 10^{-8}} = 46.86 \quad (16)$$

4. Friction factor (f_{dry}):

$$f_{dry} = \frac{2\Delta P \varepsilon^3 \rho_G}{G^2 Z a} = \frac{2 \cdot 9.81 \cdot 0.586^3 \cdot 1.1305}{0.0847^2 \cdot 0.42 \cdot 375} = 4.02 \quad (17)$$

5. Flooding parameters (π_1, π_2): For $L = 0.2 \text{ gal/min}$ and $G^* = 100\%$:

$$v = G^*/\rho_G = 0.8471/1.1305 = 0.7493 \text{ m/s} \quad (18)$$

$$\pi_1 = \left(\frac{f_{dry} a}{\varepsilon^3} \right) \frac{v^2}{2g} \cdot \frac{\rho_G}{\rho_L} \cdot \mu_{td}^{0.2} = \left(\frac{4.19 \cdot 375}{0.586^3} \right) \frac{0.7493^2}{2 \cdot 9.81} \cdot \frac{1.1305}{995.6} \cdot 1^{0.2} = 0.2583 \quad (19)$$

$$\pi_2 = \frac{L}{G} \sqrt{\frac{\rho_G}{\rho_L}} = \frac{0.0027}{0.8471} \sqrt{\frac{1.1305}{995.6}} = 0.0952 \quad (20)$$

8 REFERENCES

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